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Fluid connector including a retaining clip cartridge

Abstract

A cartridge for a fluid connector including a connector body having a bore and a flange, and a tube end form, the cartridge including a first end, a second end, a through-bore, a first radially outward facing surface including at least one aperture extending radially inward to the through-bore, and a retaining clip including at least one protrusion operatively arranged to extend into the through-bore.

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Background/Summary

FIELD

(1) The present disclosure relates to fluid connectors, and, more particularly, to a fluid connector including a cartridge for a retaining clip that eliminates the need for post-process machining of a connector body and ensures that the retaining clip is always contained.

BACKGROUND

(2) Fluid connectors are integral components for many applications, and especially for automotive applications. Since an automotive system is made up of various components such as a radiator, transmission, and engine, fluid must be able to travel not only within each component but also between components. An example of fluid traveling between components is the transmission fluid traveling from the transmission to the transmission oil cooler in order to lower the temperature of the transmission fluid. Fluid predominantly moves between components via flexible or rigid hoses which connect to each component by fluid connectors. Such fluid connectors typically include a retaining clip, retaining ring clip, or snap ring carried on the connector body which is adapted to snap behind a raised shoulder of a tube end form when the tube end form is fully inserted into the connector body. However, in order for the fluid connector to properly function, slots or apertures must be machined in the connector body such that the retaining clip can protrude therethough and engage the tube end form, which requires extra post-process manufacturing. Additionally, during the assembly process, installation of the retaining clip onto the connector body is difficult and failure to install the retaining clip properly can jeopardize the structural integrity of the retaining clip. Furthermore, since the retaining clips are very thin and small, it is easy to lose them if dropped or misplaced.

(3) Thus, there has been a long-felt need for a fluid connector including a retaining clip cartridge that eliminates the need for post-process machining, contains the retaining clip, and reduces the insertion force required to assemble the fluid connector.

SUMMARY

(4) According to aspects illustrated herein, there is provided a cartridge for a fluid connector including a connector body having a bore and a flange, and a tube end form, the cartridge comprising a first end, a second end, a through-bore, a first radially outward facing surface including at least one aperture extending radially inward to the through-bore, and a retaining clip including at least one protrusion operatively arranged to extend into the through-bore.

(5) According to aspects illustrated herein, there is provided a fluid connector, comprising a connector body, including a first through-bore, a first groove arranged circumferentially within the first through-bore, a seal arranged within the first groove, and a flange, and a cartridge operatively arranged to be inserted into the first through-bore, including a first end, a second end, a second through-bore, a first radially outward facing surface including at least one aperture extending radially inward to the second through-bore, and a retaining clip operatively arranged on the radially outward facing surface, the retaining clip including at least one protrusion extending through the at least one aperture and into the second through-bore.

(6) According to aspects illustrated herein, there is provided a fluid connector, comprising a connector body, including a first through-bore, a first groove arranged circumferentially within the first through-bore, a seal arranged within the first groove, and a malleable flange, a cartridge operatively arranged to be secured in the first through-bore, including a first end, a second end, a second through-bore, a first radially outward facing surface including at least one aperture extending radially inward to the second through-bore, and a retaining clip operatively arranged to extend through the at least one aperture and into the second through-bore, and a tube end form operatively arranged to be secured in the connector body via the cartridge.

(7) According to aspects illustrated herein, there is provided a fluid connector having a connector body, a cartridge including a retaining clip, and a tube end form. The use of a retaining clip cartridge addresses space efficiency requirements and ergonomic concerns at assembly plants. In addition, it improves component assembly and manufacturability through process improvements along with reducing debris for improved cleanliness. Manufacturer and assembler cleanliness requirements are becoming more stringent and the tube to connector body insertion force requirements are decreasing to allow for ease of assembly from all assembly positions.
(8) The clip cartridge includes molded retainer clip slots into an injection molded (e.g., powder injection molding (PIM) or metal injection molding (MIM)) cartridge, which eliminates the need for post process techniques that is typically required to make the slots during a machining process. The post-process machining leads to debris (i.e., shavings). The cartridge includes indexing slots molded therein that serve as an alignment feature for the assembly of the clip. The cartridge may comprise, for example, a polymer. The interaction of the clip and the cartridge (e.g., polymer) reduces tube insertion forces. The present disclosure ensures that the retaining clip is always contained, which addresses a quality issue that an end user may receive a fluid connector without retaining clips.
(9) The clip cartridge, when inserted and secured in the connection body, may create the second half of the seal or O-ring gland. This eliminates the need to produce an interior groove that contains the seal within the connector body and thus simplifies the manufacturing process of the connector body. Once the cartridge is inserted into the connector body, a flange of the connector body that overhangs the cartridge is crimped, swaged, folded over the proximal end of the cartridge securing it within the connector body.
(10) In some embodiments, the cartridge fully encloses the retaining clip. The retaining clip is assembled into the cartridge. A seal or an O-ring is inserted into the connector body. The cartridge and retaining clip assembly is inserted into the connector body. At the proximal end of the connector body there is an outer diameter material, flange, or returns that is crimped, swaged, folded over, etc. to secure the cartridge and retaining clip assembly therein.
(11) In some embodiments, the cartridge does not fully enclose the retaining clip and allows for a top down assembly, which reduces or eliminates the chance of radial expansion of the retaining clip during installation. A seal or O-ring is inserted into the connector body. The cartridge is inserted into the fitting. The retainer clip is placed onto the proximal end of the cartridge. At the proximal end of the connector body there is an outer diameter material, flange, or returns that is crimped, swaged, folded over, etc. to secure the cartridge and retaining clip therein. The folded material retains the clip.
(12) In some embodiments, any style retainer clip suitable for securing the tube end form in the connector body may be used. When the flange is crimped over the cartridge a retaining clip groove is formed and the cartridge and the retaining clip are secured within the connector body.
(13) These and other objects, features, and advantages of the present disclosure will become readily apparent upon a review of the following detailed description of the disclosure, in view of the drawings and appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Various embodiments are disclosed, by way of example only, with reference to the accompanying schematic drawings in which corresponding reference symbols indicate corresponding parts, in which:
(2) FIG. **1** is a perspective view of a fluid connector, in an unsecured state;
(3) FIG. **2** is a perspective view of the fluid connector shown in FIG. **1**, in a secured state;

(4) FIG. **3** is an exploded view of the fluid connector shown in FIG. **1**;
(5) FIG. **4** is a cross-sectional view of the fluid connector taken generally along line **4-4** in FIG. **1**;
(6) FIG. **5** is a cross-sectional view of the fluid connector taken generally along line **5-5** in FIG. **2**;
(7) FIG. **6** is a perspective view of a cartridge and a retaining clip;
(8) FIG. **7** is an exploded view of the cartridge and retaining clip shown in FIG. **6**;
(9) FIG. **8** is a perspective view of a fluid connector, in an unsecured state;
(10) FIG. **9** is a perspective view of the fluid connector shown in FIG. **8**, in a secured state;
(11) FIG. **10** is an exploded view of the fluid connector shown in FIG. **8**;
(12) FIG. **11** is a cross-sectional view of the fluid connector taken generally along line **11-11** in FIG. **8**;
(13) FIG. **12** is a cross-sectional view of the fluid connector taken generally along line **12-12** in FIG. **9**;
(14) FIG. **13** is a perspective view of a cartridge and a retaining clip; and,
(15) FIG. **14** is an exploded view of the cartridge and retaining clip shown in FIG. **13**.

DETAILED DESCRIPTION

(16) At the outset, it should be appreciated that like drawing numbers on different drawing views identify identical, or functionally similar, structural elements. It is to be understood that the claims are not limited to the disclosed aspects.

(17) Furthermore, it is understood that this disclosure is not limited to the particular methodology, materials and modifications described and as such may, of course, vary. It is also understood that the terminology used herein is for the purpose of describing particular aspects only, and is not intended to limit the scope of the claims.

(18) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure pertains. It should be understood that any methods, devices or materials similar or equivalent to those described herein can be used in the practice or testing of the example embodiments.

(19) It should be appreciated that the term “substantially” is synonymous with terms such as “nearly,” “very nearly,” “about,” “approximately,” “around,” “bordering on,” “close to,” “essentially,” “in the neighborhood of,” “in the vicinity of,” etc., and such terms may be used interchangeably as appearing in the specification and claims. It should be appreciated that the term “proximate” is synonymous with terms such as “nearby,” “close,” “adjacent,” “neighboring,” “immediate,” “adjoining,” etc., and such terms may be used interchangeably as appearing in the specification and claims. The term “approximately” is intended to mean values within ten percent of the specified value.

(20) It should be understood that use of “or” in the present application is with respect to a “non-exclusive” arrangement, unless stated otherwise. For example, when saying that “item x is A or B,” it is understood that this can mean one of the following: (1) item x is only one or the other of A and B; (2) item x is both A and B. Alternately stated, the word “or” is not used to define an “exclusive or” arrangement. For example, an “exclusive or” arrangement for the statement “item x is A or B” would require that x can be only one of A and B. Furthermore, as used herein, “and/or” is intended to mean a grammatical conjunction used to indicate that one or more of the elements or conditions recited may be included or occur. For example, a device comprising a first element, a second element and/or a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or, a device comprising a second element and a third element.

(21) Moreover, as used herein, the phrases “comprises at least one of” and “comprising at least one of” in combination with a system or element is intended to mean that the system or element includes one or more of the elements listed after the phrase. For example, a device comprising at

least one of: a first element; a second element; and, a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or, a device comprising a second element and a third element. A similar interpretation is intended when the phrase "used in at least one of:" is used herein. Furthermore, as used herein, "and/or" is intended to mean a grammatical conjunction used to indicate that one or more of the elements or conditions recited may be included or occur. For example, a device comprising a first element, a second element and/or a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or, a device comprising a second element and a third element.
(22) Adverting now to the figures, FIG. **1** is a perspective view of fluid connector **10**, in an unsecured state. FIG. **2** is a perspective view of fluid connector **10**, in a secured state. FIG. **3** is an exploded view of fluid connector **10**. FIG. **4** is a cross-sectional view of fluid connector **10** taken generally along line **4-4** in FIG. **1**. FIG. **5** is a cross-sectional view of fluid connector **10** taken generally along line **5-5** in FIG. **2**. Fluid connector **10** generally comprises cartridge **20**, connector body **40**, and tube end form **80**. For the purposes of the present disclosure, "unsecured state" is intended to mean that flange **52** of connector body **40** has not yet been crimped over cartridge **20**. "Secured state" is intended to mean that section **56** of flange **52** has been crimped over cartridge **20** (or cartridge **30**) thereby securing cartridge **20** and retaining clip **70** (or **170**) within connector body **40**. The following description should be read in view of FIGS. **1-5**.
(23) Connector body **40** comprises through-bore **41** extending from end **42** to end **44**, radially inward facing surface **46**, radially inward facing surface **48**, groove **50**, flange **52**, head **58**, and radially outward facing surface **60**. Connector body **40** is arranged to be connected to a component that is filled with a fluid. For example, connector body **40** may be connected to a transmission via radially outward facing surface **60**, which may comprise external threading. Connector body **40** may be screwed into a threaded hole in the transmission via head **58** (e.g., using a wrench), which is then filled with transmission fluid. In some embodiments, head **58** is hexagonal; however, it should be appreciated that head **58** may comprise any geometry suitable for applying torque to connector body **40**. Another component in which fluid connector **10**, specifically connector body **40**, may be installed into is an engine block. It should be appreciated that fluid connector **10** may be used in various other components, assemblies, and subassemblies in which fluid connection is desired. Radially outward facing surface **60** may further comprise groove **64**. Seal or O-ring **64** may be arranged in groove **64** to create a fluid tight seal between connector body **40** and the component it is connected to. Seal **62** is arranged in connector body **40**. Specifically, seal **62** is arranged in groove **50**. In some embodiments, seal **62** is an O-ring. Body **40** further comprises surface **47**, which extends between and is connected to radially inward facing surface **46** and radially inward facing surface **48**. Surface **47** is operatively arranged to engage cartridge **20**, as will be described in greater detail below. Flange **52** extends from head **58** in axial direction AD**2**. Flange **52** comprises section **54** and section **56**, which are shown in the figures as being separated by fold line L**1**. It should be appreciated that fold line L**1** is an imaginary line used solely for the purposes of describing the folding action of flange **52** herein. In some embodiments, connector body **40** comprises a metal. In some embodiments, connector body **40** comprises a polymer with flange **52** comprising a suitable malleable material. In some embodiments, connector body **40** comprises a ceramic with flange **52** comprising a suitable malleable material.
(24) Tube end form **80** comprises end **82**, section **83**, shoulder **87**, section **89**, end **94**, and through-bore **96**. Through-bore **96** extends through tube end form **80** from end **82** to end **94**. Section **83** is

arranged between end **82** and shoulder **87** and comprises radially outward facing surface **84**. Radially outward facing surface **84** includes a substantially constant diameter. In some embodiments, radially outward facing surface **84** comprises a frusto-conical taper proximate end **82** (see FIGS. **4-5**). Shoulder **87** is arranged between section **83** and section **89** and comprises radially outward facing surface **86**, radially outward facing surface **90**, and surface **88**. As shown, radially outward facing surface **86** is a frusto-conical surface extending from radially outward surface **84** to radially outward facing surface **90**. Radially outward facing surface **86** increases in diameter is axial direction AD**2**. In some embodiments, radially outward facing surface **86** is an axial surface facing at least partially in axial direction AD**1**. Radially outward facing surface **90** extends from radially outward facing surface **86** to shoulder surface **88**. In some embodiments, radially outward facing surface **90** comprises a constant diameter. In some embodiments, radially outward facing surface **90** comprises a variable diameter. Shoulder surface **88** is an axial surface facing at least partially in axial direction AD**2**. Section **89** is arranged between shoulder **87** and end **94** and comprises radially outward facing surface **92**. Radially outward facing surface **92** includes a substantially constant diameter. Tube end form **80** is arranged to be inserted, specifically with end **82** first, into connector body **40** and cartridge **20**. Tube end form **80** is inserted into connector body **40** and cartridge **20** until retaining clip **70** snaps over shoulder **87**. It is the engagement of protrusions **72**A-B with shoulder surface **88** that secures tube end form **80** within connector body **40**. It should be appreciated that tube end form **80** may be any traditional tube end form comprising a bead, radially outward extending protrusion or flange, or ramp profile, which extends radially outward and axially on the outer surface of the tube end form, to displace a retaining ring, snap ring, or wire clip within the connector body (and cartridge) to secure the tube end form within the connector body. In some embodiments, tube end form **80** comprises a metal. In some embodiments, tube end form **80** comprises a polymer. In some embodiments, tube end form **80** comprises a ceramic.

(25) Cartridge **20** is operatively arranged to carry and enclose retaining clip **70**. Cartridge **20** comprises end **22**, radially outward facing surface **24**, radially outward facing surface **30**, end **32**, and through-bore **34**. Radially outward facing surface **24** comprises radially outward facing groove **26**. Groove **26** further comprises one or more apertures (e.g., apertures **28**A-B) which extend from groove **26** to through-bore **34**. In some embodiments, the dimeter of radially outward facing surface **24** is less than the diameter of radially outward facing surface **30**. In some embodiments, the dimeter of radially outward facing surface **24** is equal to the diameter of radially outward facing surface **30**. In some embodiments, the dimeter of radially outward facing surface **24** is greater than the diameter of radially outward facing surface **30**. Cartridge **20** further comprises radially inward facing surface **36** proximate end **32** and radially inward facing surface **38** proximal end **22**. In some embodiments, radially inward facing surface **36** comprises a constant diameter. In some embodiments, radially inward facing surface **36** comprises a variable diameter. In some embodiments, radially inward facing surface **38** comprises a variable diameter (e.g., is frusto-conical). In some embodiments, radially inward facing surface **38** comprises a constant diameter with a flange arranged to engage shoulder **87**. Cartridge **20** is operatively arranged to be inserted, with end **22** first, into through-bore **41** at end **44** of connector body **40** and secured therein. In some embodiments, cartridge **20** comprises a metal. In some embodiments, cartridge **20** comprises a polymer. In some embodiments, cartridge **20** comprises a ceramic.

(26) Retaining clip (or retaining ring or snap clip/ring) **70** is arranged in groove **26** in cartridge **20**. Retaining clip **70** is generally a retaining ring including one or more protrusions extending radially inward. In the embodiment shown, retaining clip **70** comprises protrusions **72**A-B. Protrusions **72**A-B extend radially inward through apertures **28**A-B in groove **26**. Protrusions **72**A-B are arranged to engage shoulder **87**, specifically, shoulder surface **88**, to secure tube end form **80** within cartridge **20** and thus connector body **40**. Retaining clip **70** may comprise any material that is capable of elastically deforming and returning to its original shape (e.g., metal, polymer, etc.).

(27) To assembly fluid connector **10**, cartridge **20** with retaining clip **70** is inserted into connector body **40**. As shown in FIGS. **4-5**, end **22** of cartridge **20** engages surface **47** of connector body **40**. End **22** thereby forms the second half of the seal or O-ring gland by enclosing seal **62** within groove **50**. Radially outward facing surface **24** is arranged proximate to or engages with radially inward facing surface **46**. Radially outward facing surface **30** is arranged proximate to or engages with flange **52**. Flange **52** is then crimped radially inward to secure cartridge **20** within connector body **40**, as shown in FIG. **5**. Specifically, section **56** is crimped or bent radially inward about bend line L**1**, until section **56** is arranged proximate to or abuts against end **32** of cartridge **20**. In the secured state as shown in FIG. **5**, cartridge **20** is prevented from displacement in axial direction AD**1** by surface **47** and axial direction AD**2** by crimped section **56**. It should be appreciated that tube end form **80** is not inserted into cartridge **20** and connector body **40** until flange **52** has been crimped. FIG. **4** shows tube end form **80** inserted into cartridge **20** and connector body **40** in an unconnected state only to further illustrate the interaction and orientation of the components. However, tube end form **80** would not be present during the assembly of cartridge **20** and connector body **40**.
(28) FIG. **6** is a perspective view of cartridge **120** and a retaining clip **170**. FIG. **7** is an exploded view of cartridge **120** and retaining clip **170**. The following description should be read in view of FIGS. **6** and **7**.
(29) Cartridge **120** is operatively arranged to carry and enclose retaining clip **170**. Cartridge **120** comprises end **122**, radially outward facing surface **124**, radially outward facing surface **130**, end **132**, and through-bore **134**. Radially outward facing surface **124** comprises radially outward facing groove **126**. Groove **126** further comprises one or more apertures (e.g., apertures **128**A-C) which extend from groove **126** to through-bore **134**. In some embodiments, the dimeter of radially outward facing surface **124** is less than the diameter of radially outward facing surface **130**. In some embodiments, the dimeter of radially outward facing surface **124** is equal to the diameter of radially outward facing surface **130**. In some embodiments, the dimeter of radially outward facing surface **124** is greater than the diameter of radially outward facing surface **130**. Cartridge **120** further comprises radially inward facing surface **136** proximate end **132** and radially inward facing surface **138** proximal end **122**. In some embodiments, radially inward facing surface **136** comprises a constant diameter. In some embodiments, radially inward facing surface **136** comprises a variable diameter. In some embodiments, radially inward facing surface **138** comprises a variable diameter (e.g., is frusto-conical). In some embodiments, radially inward facing surface **138** comprises a constant diameter with a flange arranged to engage shoulder **87**. Cartridge **120** is operatively arranged to be inserted, with end **122** first, into through-bore **41** at end **44** of connector body **40** and secured therein. In some embodiments, cartridge **120** comprises a metal. In some embodiments, cartridge **120** comprises a polymer. In some embodiments, cartridge **120** comprises a ceramic.
(30) Retaining clip (or retaining ring or snap clip/ring) **170** is arranged in groove **126** in cartridge **120**. Retaining clip **170** is generally a retaining ring including one or more protrusions extending radially inward. In the embodiment shown, retaining clip **170** comprises protrusions **172**A-C. Protrusions **72**A-C extend radially inward through apertures **128**A-C in groove **126**. Protrusions **172**A-C are arranged to engage shoulder **87**, specifically, shoulder surface **88**, to secure tube end form **80** within cartridge **120** and thus connector body **40**. Retaining clip **170** may comprise any material that is capable of elastically deforming and returning to its original shape (e.g., metal, polymer, etc.).
(31) To assembly fluid connector **10**, cartridge **120** with retaining clip **170** is inserted into connector body **40**. End **122** of cartridge **120** engages surface **47** of connector body **40**. End **122** thereby forms the second half of the seal or O-ring gland by enclosing seal **62** within groove **50**. Radially outward facing surface **124** is arranged proximate to or engages with radially inward facing surface **46**. Radially outward facing surface **130** is arranged proximate to or engages with flange **52**. Flange **52** is then crimped radially inward to secure cartridge **20** within connector body

40. Specifically, section **56** is crimped or bent radially inward about bend line L**1**, until section **56** is arranged proximate to or abuts against end **132** of cartridge **120**. In the secured state, cartridge **120** is prevented from displacement in axial direction AD**1** by surface **47** and axial direction AD**2** by crimped section **56**.

(32) FIG. **8** is a perspective view of fluid connector **210**, in an unsecured state. FIG. **9** is a perspective view of fluid connector **210**, in a secured state. FIG. **10** is an exploded view of fluid connector **210**. FIG. **11** is a cross-sectional view of fluid connector **210** taken generally along line **11-11** in FIG. **8**. FIG. **12** is a cross-sectional view of fluid connector **210** taken generally along line **12-12** in FIG. **9**. Fluid connector **210** generally comprises cartridge **220**, connector body **240**, and tube end form **280**. In some embodiments, fluid connector **210** further comprises washer **300**. For the purposes of the present disclosure, "unsecured state" is intended to mean that flange **252** of connector body **240** has not yet been crimped over cartridge **220**. "Secured state" is intended to mean that section **256** of flange **252** has been crimped over cartridge **220** (or cartridge **320**) thereby securing cartridge **220** and retaining clip **70** (or **170**) within connector body **240**. The following description should be read in view of FIGS. **8-12**.

(33) Connector body **240** comprises through-bore **241** extending from end **242** to end **244**, radially inward facing surface **246**, radially inward facing surface **248**, groove **250**, flange **252**, head **258**, and radially outward facing surface **260**. Connector body **240** is arranged to be connected to a component that is filled with a fluid. For example, connector body **240** may be connected to a transmission via radially outward facing surface **260**, which may comprise external threading. Connector body **240** may be screwed into a threaded hole in the transmission via head **258** (e.g., using a wrench), which is then filled with transmission fluid. In some embodiments, head **258** is hexagonal; however, it should be appreciated that head **258** may comprise any geometry suitable for applying torque to connector body **240**. Another component in which fluid connector **210**, specifically connector body **240**, may be installed into is an engine block. It should be appreciated that fluid connector **210** may be used in various other components, assemblies, and subassemblies in which fluid connection is desired. Radially outward facing surface **260** may further comprise groove **264**. Seal or O-ring **626** may be arranged in groove **626** to create a fluid tight seal between connector body **424** and the component it is connected to. Seal **626** is arranged in connector body **424**. Specifically, seal **626** is arranged in groove **525**. In some embodiments, seal **626** is an O-ring. Body **424** further comprises surface **424**, which extends between and is connected to radially inward facing surface **424** and radially inward facing surface **424**. Surface **424** is operatively arranged to engage cartridge **222**, as will be described in greater detail below. Flange **252** extends from head **258** in axial direction AD**2**. Flange **252** comprises section **254** and section **256**, which are shown in the figures as being separated by fold line L**2**. It should be appreciated that fold line L**2** is an imaginary line used solely for the purposes of describing the folding action of flange **252** herein. In some embodiments, connector body **240** comprises a metal. In some embodiments, connector body **240** comprises a polymer with flange **252** comprising a suitable malleable material. In some embodiments, connector body **240** comprises a ceramic with flange **252** comprising a suitable malleable material.

(34) Tube end form **80** is arranged to be inserted, specifically with end **82** first, into connector body **240** and cartridge **220**. Tube end form **80** is inserted into connector body **240** and cartridge **220** until retaining clip **70** snaps over shoulder **87**. It is the engagement of protrusions **72**A-B with shoulder surface **88** that secures tube end form **80** within connector body **240**.

(35) Cartridge **220** is operatively arranged to carry retaining clip **70**. Cartridge **220** comprises end **222**, radially outward facing surface **224**, radially outward facing surface **230**, end **232**, and through-bore **234**. In some embodiments, radially outward facing surface **224** comprises a variable diameter (e.g., is frusto-conical or tapered) and increases in diameter is axial direction AD**2**. In some embodiments, radially outward facing surface **224** comprises a constant diameter. In some embodiments, radially outward facing surface **230** comprises a constant diameter. In some

embodiments, radially outward facing surface **230** comprises a variable diameter. Cartridge **220** further comprises one or more projections (e.g., projections **228**A-B), which extend from end **232**. Projections **228**A-B are operatively arranged to align retaining clip **70**, as will be described in greater detail below. Cartridge **220** further comprises radially inward facing surface **236** and radially inward facing surface **238**. In some embodiments, radially inward facing surface **236** comprises a constant diameter. In some embodiments, radially inward facing surface **236** comprises a variable diameter. In some embodiments, radially inward facing surface **238** comprises a variable diameter (e.g., is frusto-conical). In some embodiments, radially inward facing surface **238** comprises a constant diameter with a flange arranged to engage shoulder **87**. Cartridge **220** is operatively arranged to be inserted, with end **222** first, into through-bore **241** at end **244** of connector body **240** and secured therein. In some embodiments, cartridge **220** comprises a metal. In some embodiments, cartridge **220** comprises a polymer. In some embodiments, cartridge **220** comprises a ceramic.

(36) Retaining clip (or retaining ring or snap clip/ring) **70** is arranged proximate end **232** on projections **228**A-B. In some embodiments, retaining clip **70** is arranged on cartridge **220** such that protrusion **72**B engages the (top) space between projection **228**B and projection **228**A, and protrusion **72**A engages the (bottom) space between projection **228**B and projection **228**A. In some embodiments, retaining clip **70** is arranged on cartridge **220** such that protrusion **72**A engages the (top) space between projection **228**B and projection **228**A, and protrusion **72**B engages the (bottom) space between projection **228**B and projection **228**A. Protrusions **72**A and **72**B extend into through-bore **234** and are arranged to engage shoulder **87**, specifically, shoulder surface **88**, to secure tube end form **80** within cartridge **220** and thus connector body **240**. Retaining clip **70** may comprise any material that is capable of elastically deforming and returning to its original shape (e.g., metal, polymer, etc.).

(37) To assembly fluid connector **210**, cartridge **220** is inserted into connector body **240**. As shown in FIGS. **11-12**, end **222** of cartridge **220** engages surface **247** of connector body **240**. End **222** thereby forms the second half of the seal or O-ring gland by enclosing seal **262** within groove **250**. Radially outward facing surface **224** and radially outward facing surface **230** are arranged proximate to or engage with radially inward facing surface **246**. Retaining clip **70** is then arranged on cartridge **220**. Specifically, and as previously discussed, retaining clip **70** is arranged on projections **228**A-B such that protrusions **72**A-B extend into through-bore **234** of cartridge **220**. In some embodiments, washer **300** is then inserted into through-bore **241** of connector body **240**. As shown in FIGS. **11-12**, washer **300** is arranged proximate to or abuts against projections **228**A-B, which encloses retaining clip **70** (i.e., retaining clip **70** is prevented from displacement in axial direction AD**1** by end **232** and in axial direction AD**2** by washer **300**). Flange **252** is then crimped radially inward to secure cartridge **220** within connector body **240**, as shown in FIG. **12**. Specifically, section **256** is crimped or bent radially inward about bend line L**2**, until section **256** is arranged proximate to or abuts against washer **300**. It should be appreciated that in some embodiments, when washer **300** is not present, section **256** is crimped radially inward proximate bend line L**2** until section **256** is arranged proximate to or abuts against projections **228**A-B of cartridge **220**. In the secured state as shown in FIG. **12**, cartridge **220** is prevented from displacement in axial direction AD**1** by surface **247** and axial direction AD**2** by crimped section **256**. It should be appreciated that tube end form **80** is not inserted into cartridge **220** and connector body **240** until flange **252** has been crimped. FIG. **11** shows tube end form **80** inserted into cartridge **220** and connector body **240** in an unconnected state only to further illustrate the interaction and orientation of the components. However, tube end form **80** would not be present during the assembly of cartridge **220** and connector body **240**.

(38) FIG. **13** is a perspective view of cartridge **320** and retaining clip **170**. FIG. **14** is an exploded view of cartridge **320** and retaining clip **170**. The following description should be read in view of FIGS. **13** and **14**.

(39) Cartridge **320** is operatively arranged to carry retaining clip **170**. Cartridge **320** comprises end **322**, radially outward facing surface **324**, radially outward facing surface **330**, end **332**, and through-bore **334**. In some embodiments, radially outward facing surface **324** comprises a variable diameter (e.g., is frusto-conical or tapered) and increases in diameter is axial direction AD**2**. In some embodiments, radially outward facing surface **324** comprises a constant diameter. In some embodiments, radially outward facing surface **330** comprises a constant diameter. In some embodiments, radially outward facing surface **330** comprises a variable diameter. Cartridge **320** further comprises one or more projections (e.g., projections **328**A-C), which extend from end **332**. Projections **328**A-C are operatively arranged to align retaining clip **170**, as will be described in greater detail below. Cartridge **320** further comprises radially inward facing surface **336** and radially inward facing surface **338**. In some embodiments, radially inward facing surface **336** comprises a constant diameter. In some embodiments, radially inward facing surface **336** comprises a variable diameter. In some embodiments, radially inward facing surface **338** comprises a variable diameter (e.g., is frusto-conical). In some embodiments, radially inward facing surface **338** comprises a constant diameter with a flange arranged to engage shoulder **87**. Cartridge **320** is operatively arranged to be inserted, with end **322** first, into through-bore **241** at end **244** of connector body **240** and secured therein. In some embodiments, cartridge **320** comprises a metal. In some embodiments, cartridge **320** comprises a polymer. In some embodiments, cartridge **320** comprises a ceramic.
(40) Retaining clip (or retaining ring or snap clip/ring) **170** is arranged proximate end **332** on projections **328**A-C. In some embodiments, retaining clip **170** is arranged on cartridge **320** such that protrusion **172**A engages the space between projection **328**C and projection **328**A, protrusion **172**B engages the space between projection **228**A and projection **228**B, and protrusion **172**C engages the space between projection **228**B and projection **228**C (see FIG. **13**). However, it should be appreciated that retaining clip **170** may be arranged in any suitable orientation (i.e., protrusions **172**A-C engage any of spaces between projections **328**A-C). Protrusions **72**A-C extend into through-bore **334** and are arranged to engage shoulder **87**, specifically, shoulder surface **88**, to secure tube end form **80** within cartridge **220** and thus connector body **240**. Retaining clip **170** may comprise any material that is capable of elastically deforming and returning to its original shape (e.g., metal, polymer, etc.).
(41) To assembly fluid connector **210**, cartridge **320** is inserted into connector body **240**. End **322** of cartridge **320** engages surface **247** of connector body **240**. End **322** thereby forms the second half of the seal or O-ring gland by enclosing seal **262** within groove **250**. Radially outward facing surface **324** and radially outward facing surface **330** are arranged proximate to or engage with radially inward facing surface **246**. Retaining clip **170** is then arranged on cartridge **320**. Specifically, and as previously discussed, retaining clip **170** is arranged on projections **328**A-C such that protrusions **172**A-C extend into through-bore **334** of cartridge **320**. In some embodiments, washer **300** is then inserted into through-bore **241** of connector body **240**. Washer **300** is arranged proximate to or abuts against projections **328**A-C, which encloses retaining clip **170** (i.e., retaining clip **170** is prevented from displacement in axial direction AD**1** by end **332** and in axial direction AD**2** by washer **300**). Flange **252** is then crimped radially inward to secure cartridge **320** within connector body **240**. Specifically, section **256** is crimped or bent radially inward about bend line L**2**, until section **256** is arranged proximate to or abuts against washer **300**. It should be appreciated that in some embodiments, when washer **300** is not present, section **256** is crimped radially inward proximate bend line L**2** until section **256** is arranged proximate to or abuts against projections **328**A-C of cartridge **320**. In the secured state, cartridge **320** is prevented from displacement in axial direction AD**1** by surface **247** and axial direction AD**2** by crimped section **256**.
(42) It will be appreciated that various aspects of the disclosure above and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or

applications. Various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

REFERENCE NUMERALS

(43) **10** Fluid connector **20** Cartridge **22** End **24** Radially outward facing surface **26** Groove **28**A Aperture **28**B Aperture **30** Radially outward facing surface **32** End **34** Through-bore **36** Radially inward facing surface **38** Radially inward facing surface **40** Connector body **41** Through-bore **42** End **44** End **46** Radially inward facing surface **47** Surface **48** Radially inward facing surface **50** Groove **52** Flange **54** Section **56** Section **58** Head **60** Radially outward facing surface **62** Seal **64** Groove **66** Seal **70** Retaining clip **72**A Protrusion **72**B Protrusion **80** Tube end form **82** End **83** Section **84** Radially outward facing surface **86** Radially outward facing surface **87** Shoulder **88** Surface **89** Section **90** Radially outward facing surface **92** Radially outward facing surface **94** End **96** Through-bore **120** Cartridge **122** End **124** Radially outward facing surface **126** Groove **128**A Aperture **128**B Aperture **128**C Aperture **130** Radially outward facing surface **132** End **134** Through-bore **136** Radially inward facing surface **138** Radially inward facing surface **210** Fluid connector **220** Cartridge **222** End **224** Radially outward facing surface **228**A Projection **228**B Projection **230** Radially outward facing surface **232** End **234** Through-bore **236** Radially inward facing surface **238** Radially inward facing surface **240** Connector body **241** Through-bore **242** End **244** End **246** Radially inward facing surface **247** Surface **248** Radially inward facing surface **250** Groove **252** Flange **254** Section **256** Section **258** Head **260** Radially outward facing surface **262** Seal **264** Groove **266** Seal **320** Cartridge **322** End **324** Radially outward facing surface **328**A Projection **328**B Projection **328**C Projection **330** Radially outward facing surface **332** End **334** Through-bore **336** Radially inward facing surface **338** Radially inward facing surface L**1** Line L**2** Line AD**1** Axial direction AD**2** Axial direction

Claims

1. A cartridge for a fluid connector including a connector body having a bore and a flange, and a tube end form, the cartridge comprising: a first end; a second end; a first radially outward facing surface arranged between the first end and the second end; a through-bore; a plurality of projections extending in a first axial direction from and circumferentially spaced about the second end, the plurality of projections forming a second radially outward facing surface including at least one aperture extending radially inward to the through-bore; and a retaining clip including at least one protrusion operatively arranged to extend into the through-bore; further comprising a radially inward facing surface operatively arranged to engage a shoulder of the tube endform, wherein the radially inward facing surface is frusto-conical.
2. The cartridge as recited in claim 1, wherein the retaining clip is arranged on the second radially outward facing surface and the at least one protrusion extends through the at least one aperture.
3. The cartridge as recited in claim 1, further comprising a third radially outward facing surface extending from the first radially outward facing surface to the first end, the third radially outward facing surface being frusto-conical and decreasing in diameter in a second axial direction, opposite the first axial direction.
4. The cartridge as recited in claim 1, further comprising a washer removably engaged with the plurality of projections, wherein the retaining clip is arranged axially between the second end and the washer.
5. The cartridge as recited in claim 4, wherein: the first radially outward facing surface includes a first diameter; the second radially outward facing surface includes a second diameter; and the first diameter is greater than the second diameter.
6. The cartridge as recited in claim 1, wherein the cartridge is operatively arranged to be secured in the bore by crimping the flange radially inward.

7. A fluid connector, comprising: a connector body, including: a first end; a second end; a first through-bore; a first groove arranged circumferentially within the first through-bore; a seal arranged within the first groove; and a flange; and a cartridge arranged completely within the connector body, including: a third end; a fourth end; a second through-bore; a first radially outward facing surface including at least one aperture extending radially inward to the second through-bore; and a retaining clip operatively arranged on the first radially outward facing surface, the retaining clip including at least one protrusion extending through the at least one aperture and into the second through-bore.

8. The fluid connector as recited in claim 7, wherein the flange comprises a first section and a second section, the second section is crimped radially inward and engages the fourth end to secure the cartridge in the first through-bore and form a connected state.

9. The fluid connector as recited in claim 8, wherein: in an unconnected state, the second section is substantially aligned with the first section; and in the connected state, the second section is substantially perpendicular to the first section and is engaged with the fourth end.

10. The fluid connector as recited in claim 7, wherein the first radially outward facing surface comprises a second groove and the at least one aperture is arranged in the second groove.

11. The fluid connector as recited in claim 7, wherein the cartridge further comprises a radially inward facing surface operatively arranged to engage a shoulder of a tube end form.

12. The fluid connector as recited in claim 11, wherein the radially inward facing surface is frusto-conical.

13. The fluid connector as recited in claim 11, wherein the retaining clip is operatively arranged to secure the tube end form to the connector body.

14. The fluid connector as recited in claim 8, wherein in the connected state, the third end encloses the seal within the first groove.

15. A fluid connector, comprising: a connector body, including: a first through-bore; a first groove arranged circumferentially within the first through-bore; a seal arranged within the first groove; and a malleable flange; and a cartridge operatively arranged to be secured in the first through-bore, including: a first end; a second end; a second through-bore; a first radially outward facing surface including at least one aperture extending radially inward to the second through-bore; and a retaining clip operatively arranged to extend through the at least one aperture and into the second through-bore; wherein the malleable flange is crimped radially inward over the second end to secure the cartridge within the connector body.

16. The fluid connector as recited in claim 15, wherein the malleable flange comprises a first section and a second section, the second section is operatively arranged to be crimped radially inward to secure the cartridge in the first through-bore and form a connected state.

17. The fluid connector as recited in claim 16, wherein: in an unconnected state, the second section is not engaged with the second end; and in the connected state, the second section is engaged with the second end.

18. The fluid connector as recited in claim 7, wherein the cartridge is arranged axially between and spaced apart from the first end and the second end.